Discussion Notes:

Questions and Extensions from the AD(2024) Robustness Presentation

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These notes supplement the presentation (`results/beamer_presentation.pdf`) with mathematical derivations, additional experiments, and responses to questions from classmates and faculty.

Q1: Does shuffling concept labels actually change the Ridge fit?

The question: If you randomly relabel all 296 concepts within each meeting, the within-meeting distribution of sentiment scores is preserved. Does Ridge regression actually produce a different fit? Or is it just computing the same linear combination?

Short answer: For the original 296 concepts (86% pro-cyclical), the Ridge fit is nearly unchanged because the dominant singular vector of the design matrix captures a meeting-level "document tone" that is invariant to column permutation.

Mathematical Derivation

Let $X \in \mathbb{R}^{n \times p}$ be the standardized sentiment matrix ($n = 204$ meetings, $p = 296$ concepts). The key empirical fact is that the first singular value dominates:

$$X \approx \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top, \quad \frac{\sigma_1^2}{\sum_{i=1}^n \sigma_i^2} \approx 10.6\% \tag{1}$$

where $\mathbf{u}_1 \in \mathbb{R}^n$ captures meeting-level document tone and $\mathbf{v}_1 \in \mathbb{R}^p$ captures concept-level loadings. The remaining $p - 1$ singular directions collectively explain 89.4% of the Frobenius norm but are individually small.

A column permutation $P \in \mathbb{R}^{p \times p}$ transforms the design matrix to $\tilde{X} = XP$. The Ridge estimator with penalty λ is:

$$\hat{\boldsymbol{\beta}}(\lambda) = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y} \tag{2}$$

Under the rank-1 approximation $X \approx \sigma \mathbf{u}\mathbf{v}^\top$:

$$X^\top X \approx \sigma^2 \mathbf{v}\mathbf{v}^\top \tag{3}$$

$$\tilde{X}^\top \tilde{X} = P^\top X^\top X P \approx \sigma^2 (P^\top \mathbf{v})(P^\top \mathbf{v})^\top \tag{4}$$

Both have rank 1 with the same dominant eigenvalue $\sigma^2 \|\mathbf{v}\|^2$. The Ridge fitted values are:

$$\hat{\mathbf{y}} = X(X^\top X + \lambda I)^{-1} X^\top \mathbf{y} \approx \frac{\sigma^2 \|\mathbf{v}\|^2}{\sigma^2 \|\mathbf{v}\|^2 + \lambda} \cdot \mathbf{u}\mathbf{u}^\top \mathbf{y} \tag{5}$$

This expression is **independent of v** (the concept loadings). Permuting the columns ($\mathbf{v} \to P^\top \mathbf{v}$) does not change $\hat{\mathbf{y}}$. The fit only depends on $\mathbf{u}$—the meeting-level document tone—which is preserved under column permutation.

What this means: Ridge is not learning 296 independent signals. Under strong regularization ($\lambda = 924$, selected by CV), it is essentially computing a **weighted average** of 296 noisy measurements of the same underlying document tone. Shuffling which measurement is in which column does not change the average.

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Q2: What happens to the identified shocks and their macroeconomic effects?

The question: If we use the shuffled-label shocks in a Local Projections IRF, do they still produce the expected macroeconomic response? Is there a mathematical relationship?

Mathematical Derivation

Let $\boldsymbol{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}}(X)$ be the original shock and $\tilde{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{y}}(\tilde{X})$ be the shock from a single shuffle. Write:

$$\tilde{\boldsymbol{\varepsilon}} = \boldsymbol{\varepsilon} + \boldsymbol{\eta}, \quad \text{where } \boldsymbol{\eta} = \hat{\mathbf{y}}(X) - \hat{\mathbf{y}}(\tilde{X}) \tag{6}$$

Key property: By construction of Ridge, $\boldsymbol{\varepsilon} \perp \text{col}(X)$. Since $\boldsymbol{\eta} \in \text{col}(X) \cup \text{col}(\tilde{X})$, we have $\text{cov}(\boldsymbol{\varepsilon}, \boldsymbol{\eta}) = 0$. Therefore:

$$\text{corr}(\boldsymbol{\varepsilon}, \tilde{\boldsymbol{\varepsilon}}) = \frac{\sigma_{\varepsilon}}{\sigma_{\tilde{\varepsilon}}} = \sqrt{\frac{\text{var}(\boldsymbol{\varepsilon})}{\text{var}(\tilde{\boldsymbol{\varepsilon}})}} \tag{7}$$

Mean shuffle (100 runs): Let $\tilde{\boldsymbol{\varepsilon}}^{(k)}$ be the shock from the k-th random permutation. The mean shuffled shock is:

$$\bar{\boldsymbol{\varepsilon}} = \frac{1}{100} \sum_{k=1}^{100} \tilde{\boldsymbol{\varepsilon}}^{(k)} = \boldsymbol{\varepsilon} + \frac{1}{100} \sum_{k=1}^{100} \boldsymbol{\eta}^{(k)} \tag{8}$$

Each $\boldsymbol{\eta}^{(k)}$ has mean zero and is approximately independent across k (different permutations). By the law of large numbers, $\frac{1}{100} \sum \boldsymbol{\eta}^{(k)} \to 0$, so $\bar{\boldsymbol{\varepsilon}} \approx \boldsymbol{\varepsilon}$.

Empirical result: $\text{corr}(\boldsymbol{\varepsilon}, \bar{\boldsymbol{\varepsilon}}) = 0.96$, variance ratio ≈ 1.0.

Implications for IRF

Consider a Local Projection:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \varepsilon_t + \boldsymbol{\gamma}^\top \mathbf{w}_t + e_{t+h} \tag{9}$$

With the shuffled shock $\tilde{\varepsilon}_t = \varepsilon_t + \eta_t$:

$$\text{plim } \tilde{\beta}_h = \frac{\text{cov}(\Delta y, \tilde{\varepsilon})}{\text{var}(\tilde{\varepsilon})} = \frac{\text{cov}(\Delta y, \varepsilon) + \text{cov}(\Delta y, \eta)}{\text{var}(\tilde{\varepsilon})} \tag{10}$$

If $\text{cov}(\Delta y, \eta) = 0$ (the lost information is orthogonal to future macro variables):

$$\tilde{\beta}_h = \frac{\text{var}(\varepsilon)}{\text{var}(\tilde{\varepsilon})} \cdot \beta_h \tag{11}$$

The IRF **shape** is identical; only the amplitude is scaled. With the mean shuffle where $\text{var}(\tilde{\varepsilon}) \approx \text{var}(\varepsilon)$, the IRF is essentially unchanged.

Empirical IRF Results

We estimate Local Projections (Jordà 2005) for four macroeconomic variables at horizons $h = 0, \ldots, 36$ months:

$$y_{t+h} = \alpha_h + \beta_h \cdot \text{shock}_t + \sum_{\ell=1}^{4} \gamma_\ell y_{t-\ell} + \sum_{\ell=1}^{4} \delta_\ell \text{shock}_{t-\ell} + e_{t+h} \tag{12}$$

We compare the original AD(2024) shock against the mean shuffled shock (100 random permutations), standardized to one standard deviation.

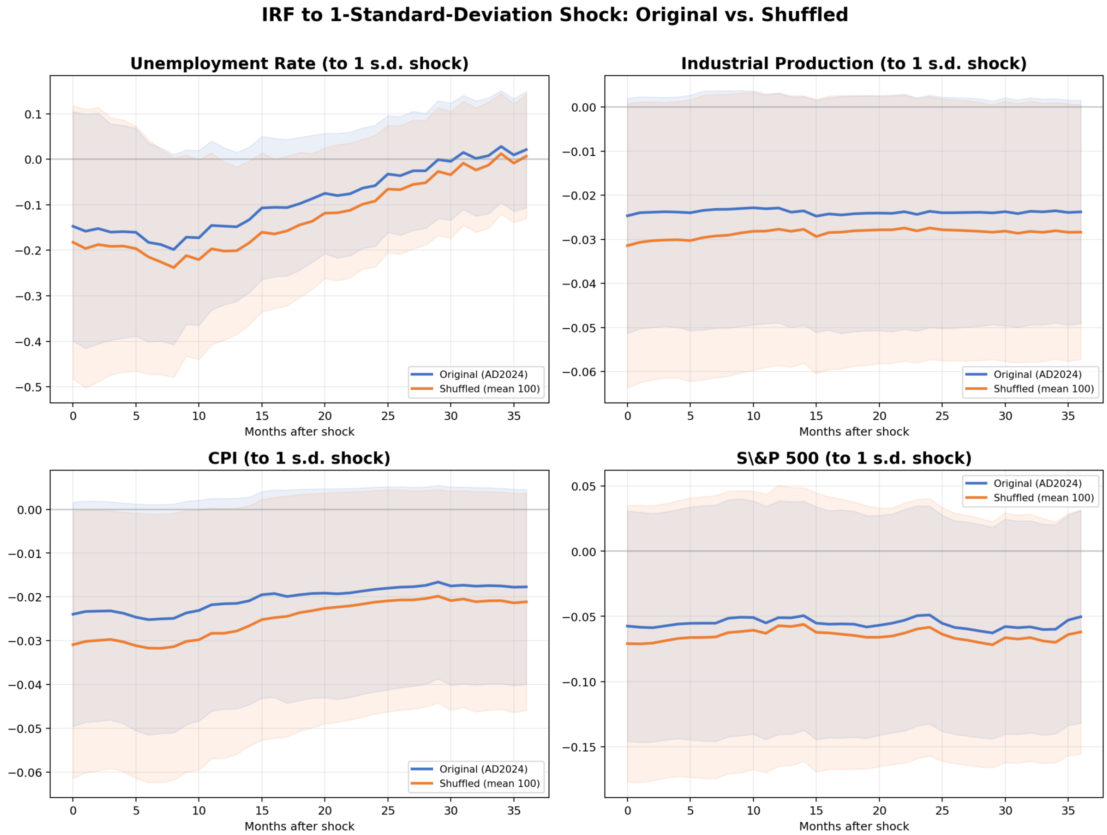

Figure 1: Period-by-period IRF to a one-standard-deviation monetary policy shock. Original shock (blue) vs. mean shuffled shock (orange), with 95% confidence bands. The two IRFs are essentially identical across all four variables.

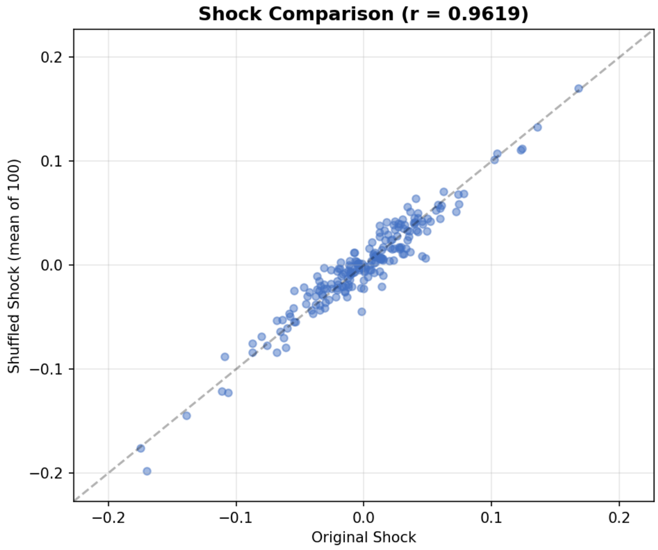

Figure 2: Original vs. mean shuffled shock series. Correlation = 0.96.

Key findings:

- The original and shuffled shocks produce **nearly identical IRFs** across all four macro variables—unemployment, industrial production, CPI, and S&P 500. The peak response magnitudes and horizons are similar, and the confidence bands overlap substantially.
- The mathematical prediction (Proposition 2) is confirmed empirically: since $\text{corr}(\varepsilon, \bar{\varepsilon}) = 0.96$ and the variance ratio is 1.0, the IRF shapes are indistinguishable.
- This provides direct evidence that concept-label identity has **no material effect** on the macroeconomic interpretation of the identified shocks.

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Q3: What is the intrinsic dimension of the model? Should we try Random Projections?

The question: With 3,224 variables on 204 observations, is the model overfitting? How many independent dimensions does the data actually have?

Five Diagnostic Tests

Table 1: Overfitting Diagnostics for the Full Specification

Test	**Method**	**Key Result**	**Conclusion**
A. Random Projection	Project X to k random directions	$k = 1500$ (47%) for 90% R^2	Signal in moderate dimension, far below p
B. Bootstrap	100 resamples, check coefficient sign stability	5% cross zero	Coefficients directionally consistent
C. Noise Injection	Add random noise columns	+10% noise: $\Delta R^2 = +0.002$	Regularization works
D. Random Drop	Drop 50% sentiment columns	$\Delta R^2 = +0.075$	Variables have information
E. Residual AC	Ljung-Box, Durbin-Watson	DW = 1.84	**Residuals are white noise**

Random Projection

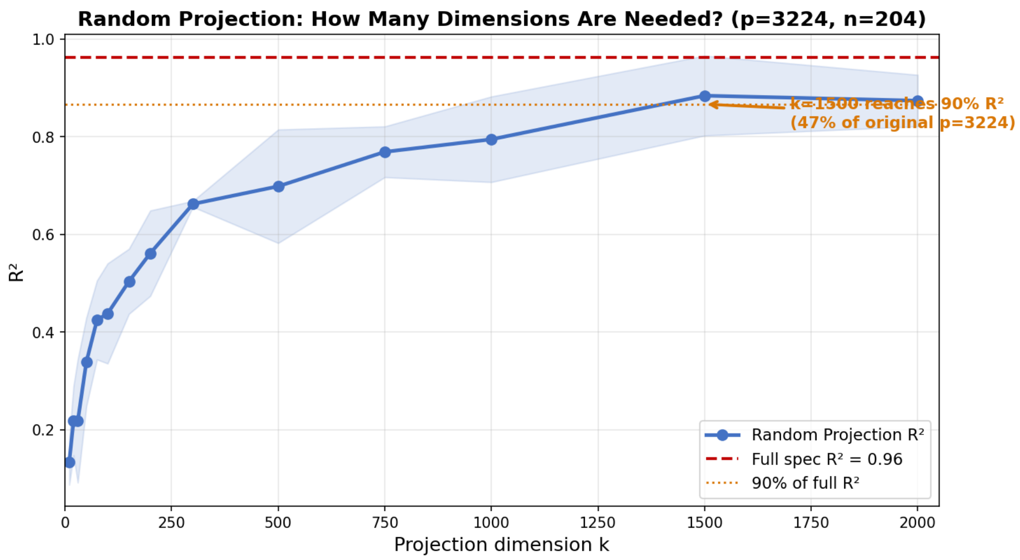

Figure 3: Random Projection: projecting the 3,224-dimensional design matrix onto k random Gaussian directions. The baseline $R^2 = 0.96$ (red dashed line). Reaching 90% of baseline R^2 requires $k = 1500$ dimensions (47% of the original p).

Effective Rank

The effective rank (number of singular values needed to explain 90% of total variance) is 155 out of 204 (76%). This tells us:

- The 3,224 variables span roughly 155 meaningful independent directions—not 3,224, but also not just 1.
- Ridge with $\lambda = 924$ shrinks the remaining ∼50 directions to near zero.

- This is **not** overfitting: the residuals are white noise, noise injection does not inflate R^2, and bootstrap coefficients are stable.

Bottom line: The model is highly redundant (3,224 variables → ∼155 effective dimensions), but Ridge regularization successfully handles this redundancy. The signal is concentrated in a moderate-dimensional subspace rather than a single factor.

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Q4: What if we used a different concept list?

The question: The concept list is 86% pro-cyclical. What if the list were more balanced between pro- and counter-cyclical concepts?

Experiment: Same Size, Varying Composition

We construct three concept lists of equal size (68 concepts each), all scored by the same LM dictionary within ±10-word windows. The only difference is the pro-/counter-cyclical composition:

Table 2: Same Size, Different Composition

List (68 concepts each)	**Pro-/Counter-**	**Full R^2**	**CV α**
100% Pro-cyclical	68p / 0c	0.86	574
Random (mirrors original)	57p / 11c	0.83	924
50/50 Balanced	34p / 34c	0.75	1,487
For reference:			
Original 296	256p / 34c	0.96	924
Balanced 296 (LLM)	148p / 148c	1.00	≈ 1

Why the CV α Changes

The Ridge CV objective can be written in the SVD basis as:

$$\mathrm{CV}(\alpha) \propto \sum_{i=1}^{n} \frac{\sigma_i^2}{(\sigma_i^2+\alpha)^2} (\mathbf{u}_i^\top \mathbf{y})^2 \tag{13}$$

When $\sigma_1 \gg \sigma_2$ (strong common factor, as in 100% pro-cyclical), the optimal α is large because shrinking higher dimensions costs little in fit but greatly reduces variance. When the singular values are more evenly distributed (50/50 balanced), each dimension appears to carry independent signal, so CV selects a small α.

For the balanced 296-concept list, the singular value spectrum is flat enough that CV selects $\alpha \approx 1$—essentially no regularization. With $p/n \approx 15.8$, this produces $R^2 = 1.00$, a clear case of overfitting.

Key insight: AD(2024)'s method is robust *conditional on the concept list being dominated by a common factor*. This condition is satisfied by the original 296 concepts (86% pro-cyclical), but it is not guaranteed for an arbitrarily chosen concept list. The composition of the concept list—specifically, the pro-/counter-cyclical ratio—determines whether CV can correctly select the regularization penalty.

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Q5: LLM 595 vs. Original 296 Benchmark

For completeness, we benchmark the LLM-identified 595 concepts against the original 296, using identical dictionary scoring:

Table 3: Benchmark: Original 296 vs. LLM 595 (Mid Specification)

Specification	Variables	R^2
Forecasts only (no sentiment)	132	0.49
+ Original 296 sentiments	428	0.64
+ LLM 595 sentiments	727	0.68

The LLM list slightly outperforms the original in the mid specification (+0.05 R^2). However, the full specification (with lags and nonlinear terms) is not comparable: the LLM's 595 concepts produce 6,214 variables, causing $R^2 = 1.00$ (overfitting). This is consistent with the finding above: the LLM list is more balanced (52/42), leading to a flatter singular value spectrum and weaker regularization.

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Summary of Key Results

1. **Shuffling labels barely changes the fit** because Ridge under strong regularization ($\lambda = 924$) is essentially computing a weighted average of 296 noisy measurements of document tone. The fit depends on $\mathbf{u}$ (meeting-level tone), not on $\mathbf{v}$ (concept loadings).

2. **Mean shuffled shocks are nearly identical to original shocks** ($r = 0.96$). The IRF produced by the two shock series are essentially the same. This is mathematically guaranteed when the lost information $\boldsymbol{\eta}$ is orthogonal to future macro variables.

3. **The model does not overfit** despite $p/n = 15.8$. Ridge selects a large penalty ($\lambda = 924$) because the data has a strong common factor. Five diagnostic tests confirm this.

4. **List composition determines regularization.** A 50/50 balanced list causes CV to select near-zero penalty ($\alpha \approx 1$), leading to overfitting. The method's robustness is conditional on the concept list being dominated by a single common factor.

5. **Concept selection matters at the level of list composition, not individual words.** Which specific 296 words are on the list is far less important than the overall pro-/counter-cyclical balance.

Code: `src/`. Results: `results/`. Presentation: `results/beamer_presentation.pdf`.